HERIOT WATT UNIVERSITY

SCHOOL OF MATHEMATICAL AND COMPUTER SCIENCES

Actuarial Mathematics and Statistics

F70LB

LIFE INSURANCE MATHEMATICS B

IN-PERSON EXAMINATION

April Examination and Assessment Diet 2025

Duration: 2 hours

Total marks available: 60

Answer all seven questions

Attached: No additional attachments
Supplied: Formulae and Tables for Examinations of the Institute and Faculty of Actuaries
Calculator Permitted: Yes

Ethical Commitment: By submitting answers to this paper, I declare that these answers are entirely my own work and have not been shared, in part or in whole or in any form, with any other student, or disseminated in any way.

I also confirm that I have reviewed and understand all University regulations, policies and procedures on assessment misconduct, including but not limited to accessing unauthorised materials during this examination or the use of unauthorised electronic devices.

I understand that infringing this statement would represent an academic misconduct offence subject to disciplinary action according to the University Regulations and Policy regarding Academic Misconduct, with possible significant consequences for degree progression and final degree outcome.

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Question 1

You are given the following information in respect of two lives (x) and (y):

$\ddot{a}_{xy} = 3$
$\ddot{a}_{x} = 6$
$\ddot{a}_{y} = 7$
${}_{10}q_{x} = 0.03$
${}_{10}q^{1}_{xy} = 0.01$
${}_{10}q_{x}{}^{1}_{y} = 0.025$
$A_{x}{}^{1}_{y} = 0.3$
$A_{x+10:\,y+10}{}^{1} = 0.1$
$A^{1}_{x+10:\,y+10} = 0.2$
Effective rate of interest $i = 2\%$ each year

Calculate the following:

(a) $a_{x|y}$

(2 marks)

(b) ${}_{10}q^{2}_{xy}$

(1 mark)

(c) ${}_{10}q_{xy}$

(1 mark)

(d) $A_{xy}{}^{1}_{\overline{:10|}}$

(2 marks)

[Total 6 marks]

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Question 2

Some UK life insurers provide loans (mortgages) to people in later life. Table 1 shows a comparison between mortality rates of these loan customers and the standard population. The column "A\E deaths" shows the "Actual" (A) mortality rate of loan customers divided by the "Expected" (E) mortality rate if they had experienced the mortality of the standard population. It is expressed as a percentage in the same way as a Standardised Mortality Ratio.

Table 1: Actual (A) vs Expected (E) mortality for loan customers

Age Band	Years since the loan began	Number of lives (thousands)	Number of deaths	Customer mortality rate	A/E deaths (%)
65-69	0	26	198	0.0076	76
70-74	0	29	269	0.0093	60
65-69	1-4	57	545	0.0096	96
70-74	1-4	78	1005	0.0129	82
65-69	5+	28	368	0.0131	131
70-74	5+	67	1167	0.0174	108
Total 65-69	-	**111**	**1111**	**0.0100**	-
Total 70-74	-	**174**	**2441**	**0.0140**	-

Table 2 shows mortality rates for the standard population.

Table 2: Standard population (females)

Age Band	Number of lives (thousands)	Mortality rate
65-69	not available	0.0100
70-74	not available	0.0155

(a) Calculate the Standardised Mortality Ratio (SMR) for all loan customers.

(2 marks)

(b) Without doing any further calculations, explain with brief reasons whether the Standardised Death Rate (SDR) for loan customers would be higher, lower or the same as the Crude Death Rate (CDR) for the standard population.

(2 marks)

(c) For each of the following types of selection, give its definition and explain with brief reasons whether there is evidence of that type of selection in the tables above.

(i) Temporary initial selection
(ii) Class selection
(iii) Adverse selection

(6 marks)

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(d)

(i) Briefly explain what a semi-Markov model is.

(2 marks)

(ii) One of your classmates suggests that it would be necessary to use a semi-Markov model, not a Markov model, for the mortality of loan customers of the kind shown in Table 1. Give brief reasons why you agree or disagree.

(2 marks)

[Total 14 marks]

Question 3

An insurer provides the following information about one of its policies and the assumptions it uses for profit-testing.

Time (t)	$q_{x+(t-1)}$	${}_{(t-1)}p_x$	Premium paid at time t-1	Profit signature σ_t
1	0.2	1.000	1000	-50
2	0.2	0.800	500	0
3	0.2	0.640	500	0
4	0.25	0.512	500	100
5	0.25	0.384	0	100

The risk discount rate is 25% a year.

Calculate each of the following:

(a) The expected present value of profit.

(b) The discounted payback period.

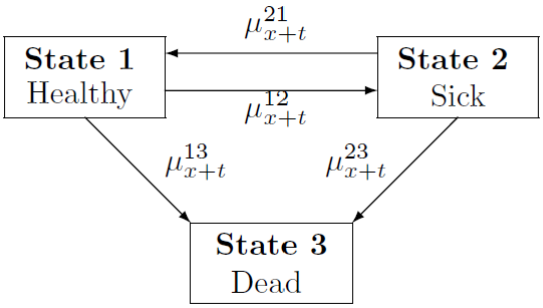
(c) The profit margin.

[Total 6 marks]

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Question 4

An insurer offers Income Protection Insurance (IPI) policies. The insurer uses the following simple multi-state model to price its policies.



(a) For each of the three states $i = 1, 2, 3$ as labelled on the diagram above, state whether you expect ${}_tp_x^{\overline{ii}} = {}_tp_x^{ii}$ or ${}_tp_x^{\overline{ii}} \neq {}_tp_x^{ii}$ and briefly why.

(2 marks)

(b) State and prove Komogorov's (forward) differential equation for $\frac{d}{dt}\left({}_tp_x^{21}\right)$.

(4 marks)

(c) Explain in words what each of the following integrals represents.

(i) $1{,}000\int_0^5 {}_tp_x^{\overline{11}}\, e^{-\delta t} dt$

(2 marks)

(ii) $20{,}000\int_0^\infty {}_tp_x^{\overline{11}}\, \mu_{x+t}^{12}\, e^{-\delta t} \int_{0.5}^\infty {}_zp_{x+t}^{\overline{22}} e^{-\delta z} dz\, dt$

(3 marks)

(d) Using the notation of part (c), write down an integral to calculate the expected present value of a lump sum of £50,000 paid on death within the first 15 years with a waiting period of one year.

(3 marks)

[Total 14 marks]

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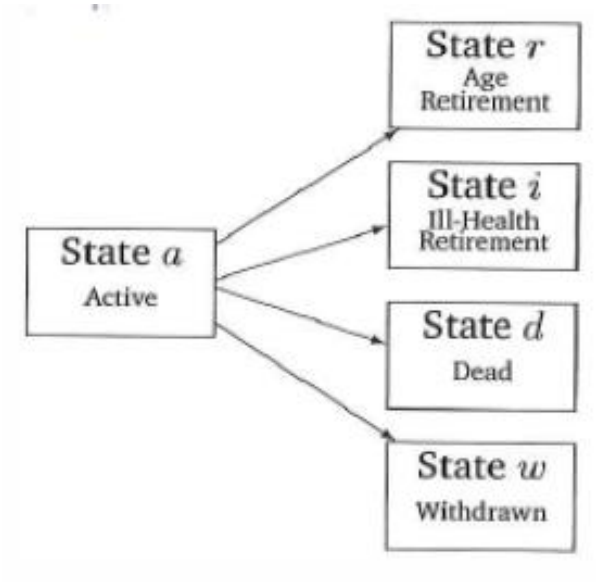
Question 5

Suppose that the survival probabilities for a pair of lives (x) and (y) are given by ${}_tp_x = e^{-10t}$ and ${}_tp_y = e^{-5t}$ respectively and their future lifetime random variables are independent. Calculate ${}_{0.1}q^{\ 1}_{x\,y}$

[Total 5 marks]

Question 6

Actuarial calculations of the expected present value of a pension will nomally use this model or something very similar:



(a) Briefly explain what is meant by each of the following:

(i) Ill-health retirement

(1 mark)

(ii) Withdrawn

(1 mark)

(b) A student says that it is not necessary to split our analysis into all the different ways of leaving and we could just have a model with two states “Active” and “Not active”. Explain with brief reasons whether or not you agree.

(2 marks)

[Total 4 marks]

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Question 7

A 25-year without-profits endowment assurance is sold to a life aged exactly 30. The policy has a sum assured of £100,000 paid at the end of the year of death or maturity. Level premiums of £4000 a year are paid quarterly.

You are given the following information in respect of a life (x):

Time (t)	Age (x+t)	A_{x+t}			${}_tp_{30}$	p_{30+t-1}
		i=3%	i=5%	i=1.942%		
25	55	0.5395	0.3775	0.6631	0.9363	0.9924
13	43	0.3997	0.2346	0.5434	0.9864	0.9982

The annual effective interest rate is 5%.

Expenses are paid by the insurer once a year at the start of the year at the same time that premiums are received. Expenses paid at time 12 (i.e. at the start of the 13th year) were £1000. It is assumed that expenses will increase by 3% (compound) every year in future.

In this question, follow the usual convention that the policy value $V(\mathrm{t})$ at time t is calculated before the payment of any premium and start-year expenses at time t and after payment of any sum assured at time t.

(a) Calculate the gross premium policy value $V(13)$ at age 43. Hence calculate the gross premium policy value $V(12.75)$ at age 42.75. State any assumptions that you use.

(10 marks)

(b) Without doing any further calculations state, with brief reasons, whether you would expect the gross premium policy value $V(13)$ to increase, decrease or stay the same if expenses at time 13 turn out to be £1010 and the policy value is immediately recalculated.

(1 mark)

[Total 11 marks]

END OF PAPER